

● RAG Learning

## B3.3 Maintaining Internal Environments

OCR A GCSE Biology (J247) Gateway — Topic B3: Organism-level systems

### 1. Homeostasis

**Homeostasis** = the maintenance of a constant internal environment despite changes in external or internal conditions. In humans, this involves controlling body temperature, blood glucose, water levels and other variables. Homeostasis usually works by **negative feedback** — a change is detected and a response counters it.

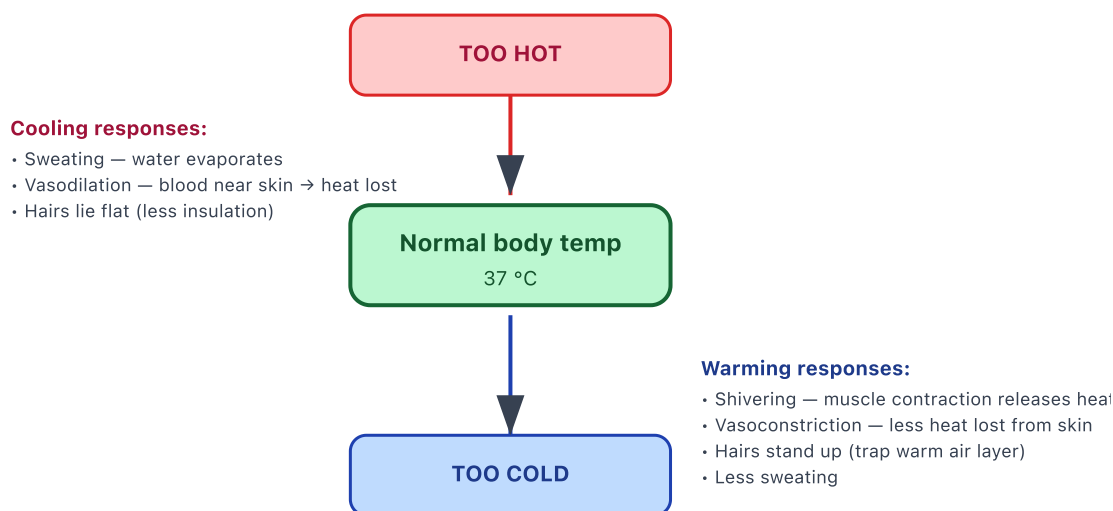
The general structure of a homeostatic control system:

**Receptor (detects change) → Coordination centre (CNS / endocrine) → Effector (muscle/gland) → Response → Negative feedback returns conditions to normal**

### 2. Thermoregulation

Body temperature is monitored by the **thermoregulatory centre** in the **hypothalamus** (in the brain). It is kept close to **37 °C** — the optimum for enzymes.

## Thermoregulation — Negative Feedback



## Mechanisms of vasodilation and vasoconstriction

| Vessel response         | When | Effect                                                                                                                               |
|-------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Vasodilation</b>     | Hot  | Arterioles supplying skin capillaries widen → more blood flows near skin → more heat lost by radiation/convection. Skin appears red. |
| <b>Vasoconstriction</b> | Cold | Arterioles supplying skin capillaries narrow → less blood flow near skin → less heat lost. Skin looks pale.                          |

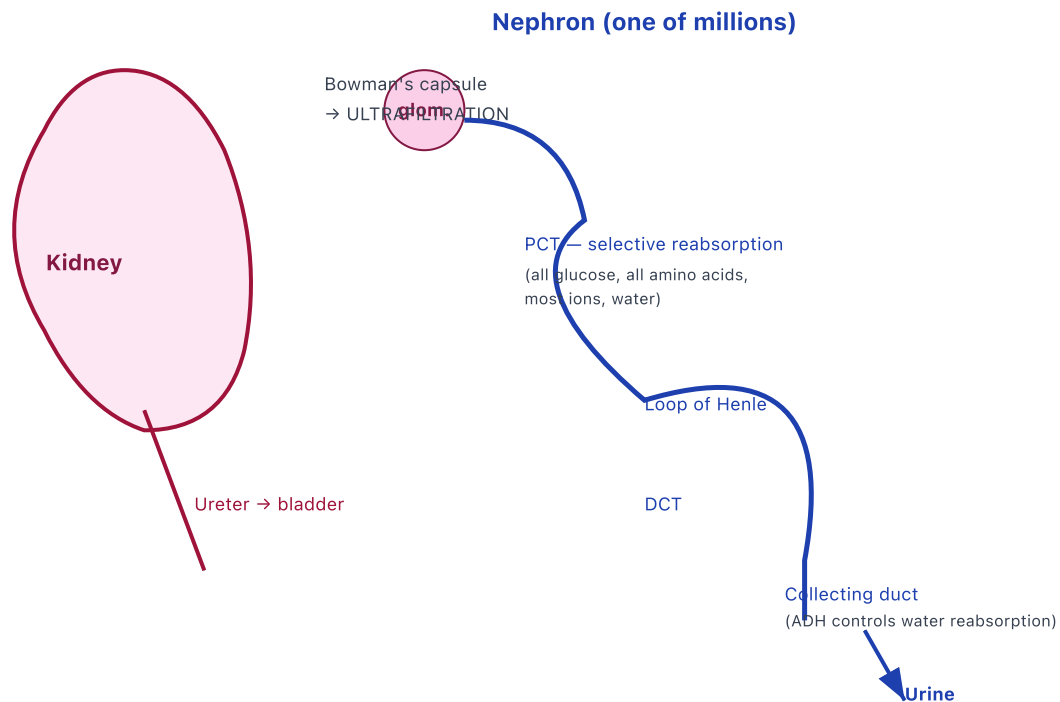
**Exam Tip:** Common misconception: capillaries themselves do NOT widen or narrow. The change happens in the **arterioles** that supply the capillary bed. Capillaries don't have muscle in their walls.

## 3. The Kidneys and Osmoregulation

**Osmoregulation** = the control of water and salt concentration in the body. The **kidneys** are the main osmoregulatory organs. They also remove **urea**, a waste product from the

breakdown of excess amino acids in the liver (deamination).

### The Kidney and a Nephron (filter unit)



### Kidney function — three steps

1. **Ultrafiltration** — at the Bowman's capsule. High-pressure blood forces small molecules (water, glucose, amino acids, salt, urea) out of the blood into the nephron. Larger items (cells, large proteins) stay in the blood.
2. **Selective reabsorption** — as the filtrate passes along the tubule:
  - **All glucose** reabsorbed back into the blood (active transport)
  - **All amino acids** reabsorbed
  - Most **ions** reabsorbed (as much as needed)
  - Most **water** reabsorbed (controlled by ADH)
3. The remaining liquid — **urine** (water, salts, urea) — passes to the bladder via the ureter.

## 4. ADH and Water Balance

**ADH (anti-diuretic hormone)** is released by the **pituitary gland**. It controls how much water is reabsorbed by the kidney tubules — adjusting urine concentration to keep blood water level constant.

| Blood is too...    | Concentrated (low water)          | Dilute (high water)         |
|--------------------|-----------------------------------|-----------------------------|
| Detected by        | Hypothalamus                      | Hypothalamus                |
| ADH from pituitary | ↑ More released                   | ↓ Less released             |
| Kidney tubule wall | More permeable to water           | Less permeable to water     |
| Water reabsorbed   | More                              | Less                        |
| Urine              | Small volume, concentrated (dark) | Large volume, dilute (pale) |

## Kidney failure

|           | Dialysis                                                                                                    | Kidney transplant                                           |
|-----------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| How       | Patient's blood passed through a machine; toxins/excess water removed across a partially permeable membrane | Donor kidney transplanted into the patient                  |
| Frequency | ~3 sessions/week, hours each                                                                                | Once (if successful)                                        |
| Pros      | Available without donor; no surgery (after fistula); reversible                                             | Cure; better quality of life                                |
| Cons      | Time-consuming; expensive long term; restricted diet                                                        | Donor shortage; immunosuppressants for life; rejection risk |

## 5. Plant Hormones and Tropisms

Plants respond to environmental stimuli by growing in particular directions. A **tropism** is a directional growth response. The main plant growth hormone is **auxin**, produced at shoot and root tips. Auxin causes shoot cells to elongate.

| Tropism                          | Response                                                                     | Adaptive value                                  |
|----------------------------------|------------------------------------------------------------------------------|-------------------------------------------------|
| <b>Phototropism</b><br>(light)   | Shoots grow TOWARDS light (positive); roots are negative or unresponsive     | Shoots get more light for photosynthesis        |
| <b>Gravitropism / Geotropism</b> | Shoots grow AWAY from gravity (negative); roots grow WITH gravity (positive) | Roots anchor and find water; shoots reach light |
| <b>Hydrotropism</b>              | Roots grow TOWARDS water                                                     | Find water for survival                         |

### Phototropism — Auxin Mechanism



## Commercial uses of plant hormones

- **Selective weed killers** — synthetic auxins kill broad-leaved weeds in lawns/cereal crops without harming the grass.
- **Rooting powders** — auxin to encourage cuttings to develop roots.
- **Tissue culture** — combinations of auxin + cytokinin grow new plants from tissue.
- **Ripening fruit** — ethene gas ripens fruit during transport (so they can be picked unripe and ripened on arrival).
- **Gibberellins** — used to end seed dormancy, promote flowering, increase fruit size.

## 6. Required Practical Link

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Linked PAGs: **investigating phototropism** in cress seedlings (light from one side); **investigating gravitropism** by orienting seeds in different directions on damp cotton wool. See PAG notes for full method.